



1. รายละเอียดของโจทย์

1.1 ชื่อโจทย์ (ต้องเป็นภาษาอังกฤษ)

Blynk blynk blynk - Malicious

1.2 หมวดของโจทย์ (Red Team, Blue Team, Miscellaneous)

Blue Team

1.3 ชื่อทีม

0xBl4ck-c4t

1.4 ระดับการแข่งขัน (มัธยมศึกษา, อุดมศึกษา)

อุดมศึกษา

1.5 รายชื่อสมาชิกในทีม

หัวหน้าทีม: -

สมาชิกคนที่ 1: -

สมาชิกคนที่ 2: -

2. รายละเอียด Write-up และการแสดงขั้นตอนการแก้ปัญหาจากโจทย์

First, I downloaded the file and checked the properties.

Command / tool: file

```
(base) (nothing@kali)-[~/Downloads]
└─$ file blynk-2-27-34.apk
blynk-2-27-34.apk: Android package (APK), with AndroidManifest.xml
```

Turns out there's a xml file that we can check so I will unpack the .apk file and then check what's inside the xml file.

Command / tool: unzip

```
(base) (nothing@kali)-[~/Downloads]
└─$ unzip blynk-2-27-34.apk
Archive:  blynk-2-27-34.apk
  inflating: AndroidManifest.xml
  inflating: classes.dex
  inflating: classes2.dex
  inflating: res/anim/abc_fade_in.xml
  inflating: res/anim/abc_fade_out.xml
  inflating: res/anim/abc_grow_fade_in_from_bottom.xml
  inflating: res/anim/abc_popup_enter.xml
  inflating: res/anim/abc_popup_exit.xml
  inflating: res/anim/abc_shrink_fade_out_from_bottom.xml
  inflating: res/anim/abc_slide_in_bottom.xml
  inflating: res/anim/abc_slide_in_top.xml
  inflating: res/anim/abc_slide_out_bottom.xml
  inflating: res/anim/abc_slide_out_top.xml
  inflating: res/anim/abc_tooltip_enter.xml
  inflating: res/anim/abc_tooltip_exit.xml
  inflating: res/anim/btn_checkbox_to_checked_box_inner_merged_animation.xml
  inflating: res/anim/btn_checkbox_to_checked_box_outer_merged_animation.xml
  inflating: res/anim/btn_checkbox_to_checked_icon_null_animation.xml
  inflating: res/anim/btn_checkbox_to_unchecked_box_inner_merged_animation.xml
  inflating: res/anim/btn_checkbox_to_unchecked_check_path_merged_animation.xml
  inflating: res/anim/btn_checkbox_to_unchecked_icon_null_animation.xml
  inflating: res/anim/btn_radio_to_off_mtrl_dot_group_animation.xml
  inflating: res/anim/btn_radio_to_off_mtrl_ring_outer_animation.xml
  inflating: res/anim/btn_radio_to_off_mtrl_ring_outer_path_animation.xml
  inflating: res/anim/btn_radio_to_on_mtrl_dot_group_animation.xml
  inflating: res/anim/btn_radio_to_on_mtrl_ring_outer_animation.xml
  inflating: res/anim/btn_radio_to_on_mtrl_ring_outer_path_animation.xml
  inflating: res/anim/design_bottom_sheet_slide_in.xml
  inflating: res/anim/design_bottom_sheet_slide_out.xml
  inflating: res/anim/design_snackbar_in.xml
  inflating: res/anim/design_snackbar_out.xml
  inflating: res/anim/fade_in.xml
  inflating: res/anim/fade_out.xml
  inflating: res/anim/fragment_fast_out_extra_slow_in.xml
  inflating: res/anim/mtrl_bottom_sheet_slide_in.xml
  inflating: res/anim/mtrl_bottom_sheet_slide_out.xml
  inflating: res/anim/mtrl_card_lowlers_interpolator.xml
  inflating: res/anim/slide_down.xml
  inflating: res/anim/slide_from_left.xml
  inflating: res/anim/slide_from_right.xml
  inflating: res/anim/slide_to_left.xml
  inflating: res/anim/slide_to_right.xml
```

Then I cat the file to read it all.

Command / tool: cat

```
(base) (nothing@kali) [~/Downloads/blynk_jadx/resources]
$ cat AndroidManifest.xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    android:versionCode="2279850"
    android:versionName="2.27.34"
    android:installLocation="auto"
    android:compileSdkVersion="31"
    android:compileSdkVersionCodename="12"
    package="cc.blynk"
    platformBuildVersionCode="31"
    platformBuildVersionName="12">
    <uses-sdk
        android:minSdkVersion="19"
        android:targetSdkVersion="32" />
    <supports-screens android:anyDensity="true" />
    <uses-permission android:name="android.permission.INTERNET" />
    <uses-permission android:name="android.permission.ACCESS_WIFI_STATE" />
    <uses-permission android:name="android.permission.CHANGE_WIFI_STATE" />
    <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />
    <uses-permission android:name="android.permission.CHANGE_NETWORK_STATE" />
    <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
    <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
    <uses-permission android:name="android.permission.ACCESS_BACKGROUND_LOCATION" />
    <uses-permission android:name="android.permission.CAMERA" />
    <uses-permission android:name="android.permission.WAKE_LOCK" />
    <uses-permission android:name="android.permission.VIBRATE" />
    <uses-permission android:name="android.permission.SCHEDULE_EXACT_ALARM" />
    <uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE" />
    <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />
    <uses-permission android:name="com.android.vending.BILLING" />
    <uses-permission android:name="android.permission.BLUETOOTH" />
    <uses-permission android:name="android.permission.BLUETOOTH_ADMIN" />
    <uses-permission android:name="android.permission.BLUETOOTH_CONNECT" />
    <uses-permission android:name="android.permission.BLUETOOTH_SCAN" />
    <uses-permission android:name="android.permission.RECEIVE_BOOT_COMPLETED" />
    <uses-permission android:name="android.intent.action.MY_PACKAGE_REPLACED" />
    <uses-permission android:name="com.android.launcher.permission.INSTALL_SHORTCUT" />
    <uses-feature
        android:name="android.hardware.sensor.light"
        android:required="false" />
    <uses-feature
        android:name="android.hardware.sensor.ambient_temperature"
        android:required="false" />
    <uses-feature
        android:name="android.hardware.sensor.proximity"
        android:required="false" />
    <uses-feature
        android:name="android.hardware.sensor.accelerometer"
        android:required="false" />
    <uses-feature
        android:name="android.hardware.bluetooth_le"
        android:required="false" />
    <uses-feature
        android:name="android.hardware.bluetooth" />
</manifest>
```

From the problem we have to find the host so I use grep to filter out all unnecessary details.

Command / tool: grep

```
(base) (nothing@kali) [~/Downloads/blynk_jadx/resources]
$ cat AndroidManifest.xml | grep "host"
    android:host="@string/app_host_token" />
    android:host="@string/app_host_restore" />
    android:host="@string/host_restore" />
    android:path="@string/app_host_restore_path" />
    android:host="@string/host_restore" />
    android:path="@string/app_host_restore_path" />
    android:host="CYBERHEROCTF{drp32k77.todoinfonet.com}" />
```

Flag: CYBERHEROCTF{drp32k77.todoinfonet.com}